

Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Junior Software Engineer with hands-on Java and Spring Boot backend experience, building REST APIs backed by JPA/Hibernate persistence, JUnit test suites, and OpenAPI-documented endpoints. Comfortable working within Scrum/Kanban teams and version-controlled workflows. Complements this with proven Golang and full-stack (React/TypeScript) delivery, Docker containerization, and CI/CD via GitHub Actions. Builds Java depth through self-directed projects, including a Kafka-based event-driven ticketing platform and a from-scratch HTTP/1.1 server, reflecting a consistent pattern of learning new stacks independently.

Core Competencies

- **Java Backend Development**: Spring Boot, REST APIs, Hibernate/JPA, OpenAPI/Swagger documentation, JUnit
- **Distributed Systems**: Docker, Apache Kafka, event-driven microservices
- **Agile Practice**: Scrum and Kanban teams, Git version control, code reviews
- **CI/CD & Cloud-Adjacent Tooling**: GitHub Actions, Docker Compose, Testcontainers
- **Additional Languages**: Golang, TypeScript/JavaScript (React), Python, Kotlin, SQL

Professional Experience

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  R&D Developer — *Feb 2026–Jun 2026*
  - Implemented LLM-based improvements to a search engine
  - Built a multi-step agent converting natural-language questions into validated, read-only SQL queries against a PostgreSQL schema, including safety validation and error-driven retries
  - Applied safety-conscious agentic design, including execution validation and retry-on-error handling, within an Agile/Scrum team workflow on a production-adjacent data-governance platform
- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  Full-Stack Developer Intern — *Feb 2024–Jun 2024*
  - Developed and maintained a business-critical corporate personnel management application (Intranet)
  - Implemented new frontend features using React and TypeScript, improving internal usability
  - Contributed to backend development, ensuring smooth integration between services via version-controlled Git workflows

Selected Projects

- **Kafka + Spring Boot Ticketing Platform**: Event-driven microservices (reservation, payment, notification, audit) built with Spring Boot 3 and Apache Kafka; outbox pattern, dead-letter topics, idempotent processing, Testcontainers integration tests.
- **English Learning API**: Spring Boot REST API for practicing English phrasal verbs and collocations; JPA entities, DTOs/mappers, Jakarta validation, centralized error handling, randomized practice challenges with attempt tracking.

- **HTTP from TCP (Java)**: Java re-implementation of an HTTP/1.1 server directly on raw sockets, built to deepen Java fundamentals: I/O, concurrency via executors, JUnit testing.

Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Master's Degree in Computer Engineering (MUIINF) *2024–2026*
  In progress (expected Sep 2026). Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  Grado (BSc) in Ingeniería Informática, Software Engineering track *2020–2024*
  GPA: 8.88/10. Thesis: "ValenMove - Unified Public Transport App" (grade 9/10).

Languages

- Spanish (native), Catalan (native), English (B2, confident conversational).

References

- Available upon request.